

About

Bioinformatics Scientist with experience in analyzing and visualizing Next Generation Sequencing (NGS) data. Strong background in translational biomedical research, with a focus on extracting meaningful biological insights from complex datasets. Experienced in developing interactive web applications, websites and tools that enhance data exploration and accessibility for researchers.

Work Experience

- **Bioinformatics Scientist**, [Bioinformatics Lab](#), 11/2023 – present
Institute of Applied Biosciences, CERTH, Thessaloniki, Greece
Supervisor: [Dr. Fotis Psomopoulos](#)
 - Participated in the organization and delivery of data analysis activities for EU-funded research projects.
 - Designed, optimized, and deployed bioinformatics workflows for genomics, transcriptomics, metagenomics and biodiversity data.
- **Master Student**, Immunogenetics Lab, 10/2021 – 10/2023
Institute of Applied Biosciences, CERTH, Thessaloniki, Greece
 - Conducted wet lab experiments and NGS workflows for research on hematological malignancies.
 - Performed bioinformatics analysis of sequencing data, integrating variant and gene-level insights.
- **Undergraduate Student**, Immunogenetics Lab, 03/2020 – 09/2021
Institute of Applied Biosciences, CERTH, Thessaloniki, Greece
 - Assisted in basic laboratory procedures and introductory bioinformatics tasks for research projects.
- **Internship**, Immunogenetics Lab, 07/2019 – 08/2019
Institute of Applied Biosciences, CERTH, Thessaloniki, Greece
 - Gained experience in fundamental molecular biology techniques and laboratory practices.

Education

- **M.Sc. in Translational Research in Biomedicine**, 2021 – 2023
Molecular Biology & Genetics Department, Democritus University of Thrace, Alexandroupoli, Greece
Supervisor: [Dr. Anastasia Chatzidimitriou](#)
- **B.Sc. in Biochemistry & Biotechnology**, 2016 – 2021
Biochemistry & Biotechnology Department, University of Thessaly, Larisa, Greece
Supervisor: [Dr. Kostas Stamatopoulos](#)

Scholarship

- 10/2021 – 10/2023, Received a scholarship from the INAB | CERTH to support my Master's research.

Technical skill

- **Languages:** R, Python, Bash, HTML, CSS
- **Data Science:** ggplot2, Quarto, R shiny, data.table, stringr, tidyverse
- **Bioinformatics Tools:** STAR, SAMTOOLS, GATK, TrimGalore, DESeq2, Kraken2
- **Workflows:** Nextflow
- **Version Control & Deployment:** Git, GitHub

Selected Funded Projects

- [LATE-AYA](#), 06/2025 – present, Help Adolescents and young adults (AYA) survivors monitor and manage late effects of cancer treatment (*European Union's Horizon Europe - HORIZON.2.1 - Health*)
- [Odyssey](#), 05/2024 – present, Connecting molecular and geographical biodiversity data (*ELIXIR BFSP commissioned service, part of the 2024-2028 Biodiversity, Food Security & Pathogens Programme*)

A comprehensive list of my participation in funded research projects is available at my [Personal page](#).

Selected Publications & Conference Activity

Publications

- Psomopoulos F, O'Cathail C, **Anastasiadou N** et al., “Toward a unified approach: Considerations for bioinformatic and sequencing activities & data in wastewater surveillance of biologic public health threats,” Open Research Europe, vol. 5, p. 267, Sep. 2025, doi: [10.12688/openreseurope.20934.1](https://doi.org/10.12688/openreseurope.20934.1).

Poster

- N. Pechlivanis, **A. Anastasiadou**, A. Papageorgiou, E. Pafilis, and F. Psomopoulos, “Odyssey: an R Shiny App Approach to explore Molecular Biodiversity in Greece, 23rd European Conference on Computational Biology (ECCB)”, Sep. 2024. doi: <https://doi.org/10.5281/zenodo.14186453>

A full portfolio of my publications and posters is available on my [Google Scholar](#) and at my [Personal page](#).

Community & Open-Source Contributions

- **Immunovisual**: Interactive gallery of Immunogenetics charts built with R ([Quarto webpage](#))
- **TidyTuesday**: Data visualization contributions ([GitHub](#) | [Shiny App](#))
- **30DayChartChallenge**: Data visualization contributions ([GitHub](#) | [Quarto webpage](#))

Courses & Certifications

- Python for Data Science and Machine Learning (Udemy)
- R programming: Advanced Analytics In R For Data Science (Udemy)
- R programming: A-Z: R for Data Science With Real Exercises (Udemy)
- Introduction to Generative AI (Google Cloud Skills Boost)
- Introduction to Large Language Models (Google Cloud Skills Boost)

Languages

- English: Full professional proficiency (C2)
- German: Intermediate proficiency (B1)
- Greek: Native proficiency